

#HW 10

● Graded

Student

Alexis Cortes

Total Points

97 / 100 pts

Question 1

(no title)

30 / 30 pts

✓ - 0 pts Correct

Question 2

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 2 pts Mass flow rate = 0.53 kg/s

- 0 pts Click here to replace this description.

- 2 pts Temperature= -10 C

Question 3

(no title)

14 / 15 pts

- 0 pts Correct

✓ - 1 pt You have to add the two to get the total power -406 kW.

- 1 pt Power -406.2 kW (calculation is wrong.)

- 1 pt Power should be in negative.

- 1 pt Mass flowrate should be 2.2 kg/s.

- 6 pts incomplete

Question 4

(no title)

14 / 15 pts

- 0 pts Correct

✓ - 1 pt The answer will be positive.

- 6 pts incomplete

Question 5

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt The unit of temperature is degrees C.
- 1 pt The temperature will be around 575 degree C.
- 1 pt Work will be around 15-17 MW
- 6 pts incomplete

Question 6

(no title)

14 / 15 pts

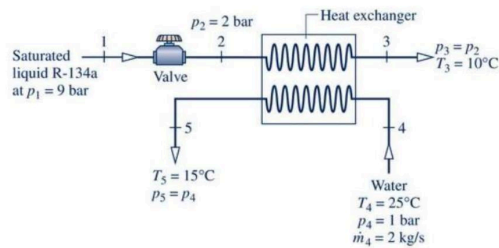
- 0 pts Correct

✓ - 1 pt The second answer should be around 1327 s.

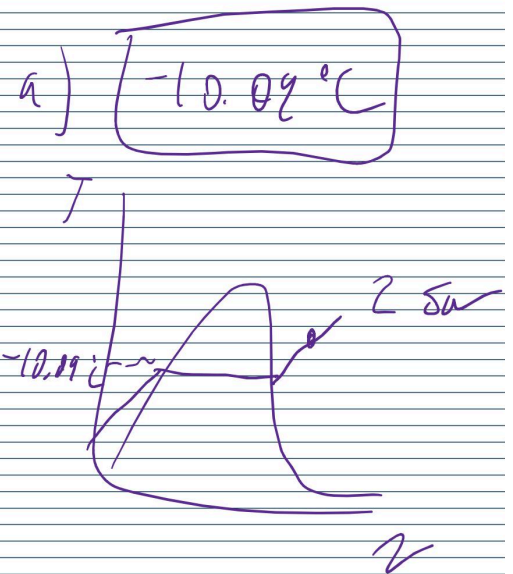
- 6 pts incomplete

Question assigned to the following page: [1](#)

The figure below provides steady-state data for a throttling valve in series with a heat exchanger. Saturated liquid Refrigerant 134a enters the valve at a pressure of 9 bar and is throttled to a pressure of 2 bar. The refrigerant then enters the heat exchanger, exiting at a temperature of 10°C with no significant decrease in pressure. In a separate stream, liquid water at 1 bar enters the heat exchanger at a temperature of 25°C with a mass flow rate of 2 kg/s and exits at 1 bar as liquid at a temperature of 15°C. Stray heat transfer and kinetic and potential energy effects can be ignored. Determine



- (a) the temperature, in °C, of the refrigerant at the exit of the valve.
 (b) the mass flow rate of the refrigerant, in kg/s.



Valve: $h_1 = h_2$

$$h_1 (9 \text{ bar, sat. liq.}) = 99.56 \frac{\text{kJ}}{\text{kg}} = h_2$$

$$\text{HEX: } \frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$$

$$0 = \dot{m}_{\text{R134A}} (h_2 - h_3) + \dot{m}_{\text{H}_2\text{O}} (h_4 - h_5)$$

$$h_3 (\text{R134A, 2 bar, 10°C}) = 258.89 \frac{\text{kJ}}{\text{kg}}$$

$$h_5 (\text{water, 1 bar, 15°C}) \approx 62.99 \frac{\text{kJ}}{\text{kg}}$$

compressed li $\approx$ sat. li

$$h_4 (\text{water, 1 bar, 25°C}) \approx 104.89 \frac{\text{kJ}}{\text{kg}}$$

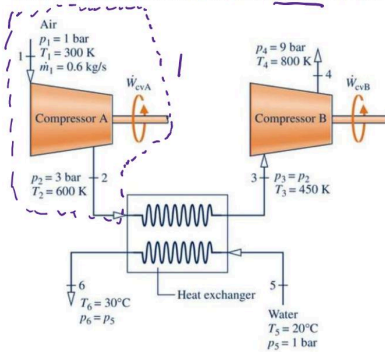
Question assigned to the following page: [1](#)

$$0 = \dot{m}_{R34A} (99.56 - 258.89) + 2 \frac{\text{kg}}{\text{s}} (104.89 - 62.29)$$

$$\dot{m}_{134A} = 0.526 \frac{\text{kg}}{\text{s}}$$

Question assigned to the following page: [2](#)

- (a) Separate streams of air and water flow through the compressor and heat exchanger arrangement shown in the figure below. Steady-state operating data are provided on the figure. Heat transfer with the surroundings can be neglected, as can all kinetic and potential energy effects. The air is modeled as an ideal gas. Determine



- a. the total power required by both compressors, in kW.
b. the mass flow rate of the water, in kg/s.

A:

$$\dot{W}_{crA} = \dot{m}_A (h_1 - h_2) = 0.6 \frac{\text{kg}}{\text{s}} (300.19 - 607.02)$$

$$h_1 (300 \text{ K}) = 300.19 \quad = -184.098 \text{ kW}$$

$$h_2 (600 \text{ K}) = 607.02 \quad \text{for A}$$

B: $\dot{W}_{crB} = \dot{m}_B (h_3 - h_4) \quad \dot{m}_A = \dot{m}_B$

$$h_3 (450 \text{ K}) = 451.8$$

$$h_4 (800 \text{ K}) = 821.95$$

$$\dot{W}_{crB} = 0.6 \frac{\text{kg}}{\text{s}} (451.8 - 821.95) = -222.09 \text{ kW}$$

Water:

$$0 = \dot{m}_{\text{water}} (h_5 - h_6) + 0.6 (h_2 - h_3) = \dot{m}_{\text{water}} (83.96 - 125.79) + 0.6 (607.02 - 451.8)$$

$$h_6 (\text{water}, 1 \text{ bar}, 30^\circ \text{C}) \approx 125.79 \quad \dot{m}_{\text{water}} = 2.23 \frac{\text{kg}}{\text{s}}$$

Question assigned to the following page: [2](#)

$$h_b(\text{water}, 1 \text{ bar}, 30^\circ\text{C}) \approx 125.79$$

$$h_f(\text{water}, 1 \text{ bar}, 20^\circ\text{C}) \approx 83.26$$

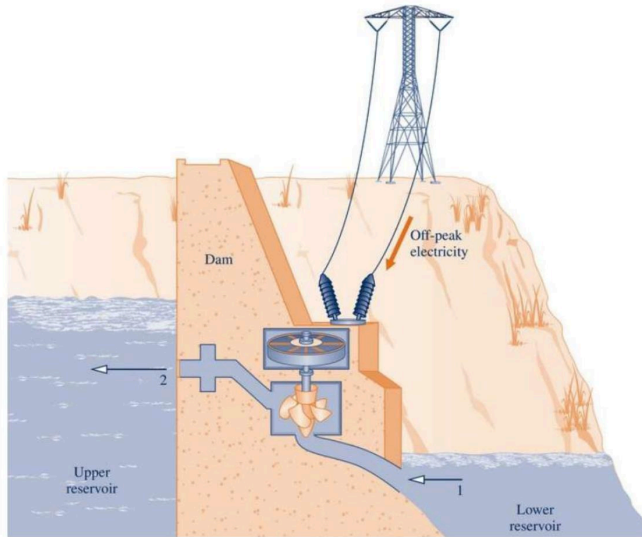
$$\dot{m}_{\text{water}} = 2.23 \frac{\text{kg}}{\text{s}}$$

Question assigned to the following page: [3](#)

7.18.72

Wednesday, November 5, 2025 2:54 PM

- (a) The figure below shows a pumped-hydro energy storage system delivering water at steady state from a lower reservoir to an upper reservoir using off-peak electricity (see subsection titled Pumped-hydro and compressed-air energy storage in Sec. 7.6). Water is delivered to the upper reservoir at a volumetric flow rate of $150 \text{ m}^3/\text{s}$ with an increase in elevation of 20 m . There is no significant change in temperature, pressure, or kinetic energy from inlet to exit. Heat transfer from the pump to its surroundings occurs at a rate of 0.6 MW and $g = 9.81 \text{ m/s}^2$. Determine the pump power required, in MW. Assuming the same volumetric flow rate when the system generates on-peak electricity using this water, will the power be greater, less, or the same as the pump power? Explain.



$$0.6 \text{ MW} = 600 \text{ kW}$$

$$\dot{m}_1 = \dot{m}_2$$

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \dot{m}_i \left(h_i + \frac{V_i^2}{2} + g z_i \right) - \dot{m}_e \left(h_e + \frac{V_e^2}{2} + g z_e \right)$$

$$\dot{W} = 600 \text{ kW} + \dot{m} \left(9.81 \frac{\text{m}}{\text{s}^2} \right) (-20 \text{ m})$$

$$\dot{V}_e = 150 \frac{\text{m}^3}{\text{s}}$$

$$\dot{m} = \frac{150 \frac{\text{m}^3}{\text{s}}}{0.001 \frac{\text{m}^3}{\text{kg}}} = 150,000 \frac{\text{kg}}{\text{s}}$$

$$\rho = 1000 \frac{\text{kg}}{\text{m}^3}$$

$$\dot{m} = 150,000 \frac{\text{kg}}{\text{s}}$$

Question assigned to the following page: [3](#)

$$V = 0.001 \frac{\text{m}^3}{\text{s}}$$

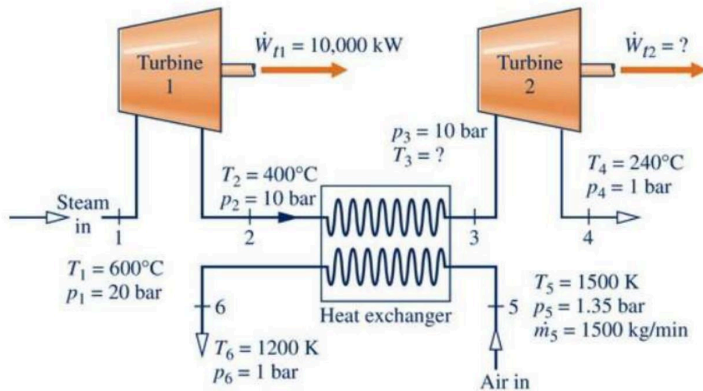
$$\dot{W} = 600 \text{ kW} - 24430000 \text{ W}$$

$$= (-28.83 \text{ MW})$$

The power generated by the on-peak electricity should be less than the power used by the pump, due to heat transfer.

Question assigned to the following page: [4](#)

- (a) Separate streams of steam and air flow through the turbine and heat exchanger arrangement shown in the figure below. Steady-state operating data are provided on the figure. Heat transfer with the surroundings can be neglected, as can all kinetic and potential energy effects. Determine (a) T_3 , in K, and (b) the power output of the second turbine, in kW.



Turbine 1:

$$10000 = \dot{m}(h_1 - h_2) = \dot{m}(3620.1 - 3263.9)$$

$$h_1 (\text{water}, 20 \text{ bar}, 600^\circ\text{C}) = 3620.1$$

$$h_2 (\text{water}, 10 \text{ bar}, 400^\circ\text{C}) = 3263.9$$

$$\dot{m}_1 = 23.76 \frac{\text{kg}}{\text{s}}$$

HEX:

$$0 = 23.76 \frac{\text{kg}}{\text{s}} (h_2 - h_3) + 25 \frac{\text{kg}}{\text{s}} (h_5 - h_6)$$

Question assigned to the following page: [4](#)

$$P_{r5} = \frac{1.35 \text{ bar}}{37.7 \text{ bar}} = 0.04 \quad T_{r5} = \frac{1500 \text{ K}}{133 \text{ K}} = 11.3$$

$$Z_{r5} \approx 1$$

$$P_{r6} = \frac{1 \text{ bar}}{37.7 \text{ bar}} = 0.03 \quad T_{r6} = \frac{1200 \text{ K}}{133 \text{ K}} = 9.02$$

$$Z_{r6} \approx 1$$

Can assume ideal gas

$$h_T(1500 \text{ K}) = 1635.97 \frac{\text{kJ}}{\text{kg}}$$

$$h_6(1200 \text{ K}) = 1277.79 \frac{\text{kJ}}{\text{kg}}$$

$$0 = 23.46(3263.9 - h_3) + 21(1635.97 - 1277.79)$$

$$h_3 = 3645.59 \frac{\text{kJ}}{\text{kg}}$$

$$\frac{600 - 540}{3647.9 - 3565.6} = \frac{T_3 - 540}{3645.59 - 3565.6}$$

$$T_3 = 576.28^\circ \text{C} = 849.43 \text{ K}$$

Question assigned to the following page: [4](#)

$$T_3 = 576.28^\circ \text{C} = 849.43 \text{K}$$

Turbine 2:

$$\dot{W}_{t2} = 23.46 \frac{\text{kg}}{\text{s}} (h_3 - h_4) = 23.46 (3675.5 - 2954.5)$$

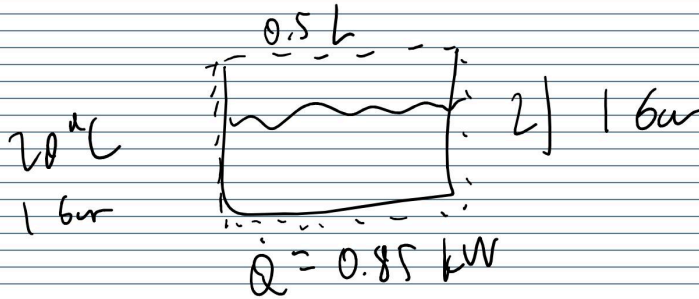
$$h_4 (\text{water}, 1 \text{ bar}, 240^\circ \text{C}) = 2954.5$$

$$\dot{W}_{t2} = 16212.97 \text{ kW}$$

Question assigned to the following page: [5](#)

An open cooking pot containing 0.5 liter of water at 20°C, 1 bar sits on a stove burner. Once the burner is turned on, the water is gradually heated at a rate of 0.85 kW while pressure remains constant. After a period of time, the water starts boiling and continues to do so until all of the water has evaporated. Determine

- (a) the time required for the onset of evaporation, in s.
 (b) the time required for all of the water to evaporate, in s, once evaporation starts.



$$\frac{dE_w}{dt} = \dot{Q} - \cancel{\dot{m}_i} + \cancel{\dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right)} - \cancel{\dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)}$$

$$\frac{dE_w}{dt} = \dot{Q} - \dot{m}_e h_e$$

$$\frac{dm_w}{dt} = \cancel{\dot{m}_i} - \dot{m}_e \quad \frac{dm_w}{dt} = -\dot{m}_e \quad \int_{m_w(0)}^{m_w(t)} dm_w = \int_0^t -\dot{m}_e dt$$

$$\frac{d(U + \cancel{KE} + \cancel{PE})_w}{dt} = \dot{Q} - \dot{m}_e h_e$$

$$\frac{dU_w}{dt} = \dot{Q} - \dot{m}_e h_e$$

$$\int_{U_w(0)}^{U_w(t)} dU_w = \int_0^t (\dot{Q} - \dot{m}_e h_e) dt$$

$$U_w(t) - U_w(0) = (\dot{Q} - \dot{m}_e h_e) t$$

$$\Delta u = \frac{(\dot{Q} - \dot{m}_e h_e) t}{-\dot{m}_e t}$$

Question assigned to the following page: [5](#)

$$U_{cv}(t) - U_{cv}(0) = -\dot{m}_e h_e t$$

$$m_{cv}(t) - m_{cv}(0) = -\dot{m}_e t$$

$$u(1 \text{ bar}, 20^\circ\text{C}) \approx 1.0018 \times 10^3 \frac{\text{J}}{\text{kg}} \quad \Delta u = \frac{\dot{Q} - \dot{m}_e h_e}{-\dot{m}_e}$$

$$\rho = 998.2 \frac{\text{kg}}{\text{m}^3}$$

$$V = 0.5 \text{ L} \cdot \frac{1 \text{ m}^3}{10^3 \text{ L}} = 5 \times 10^{-4} \text{ m}^3$$

$$m_{cv}(0) = \rho \cdot V = 0.499 \text{ kg}$$

$$u(0 \text{ sec}, 1 \text{ bar}, 20^\circ\text{C}) \approx 83.95 \frac{\text{kJ}}{\text{kg}}$$

$$u(t \text{ sec}, 1 \text{ bar}, \text{sat vapor}) = 2506.1 \frac{\text{kJ}}{\text{kg}}$$

$$h_e(t \text{ sec}, 1 \text{ bar}, \text{evap}) = 2258.0$$

$$(2506.1 - 83.95)(\dot{m}_e) = 0.45 + \dot{m}_e(2258)$$

$$\dot{m}_e = 0.0052 \frac{\text{kg}}{\text{s}}$$

Questions assigned to the following page: [5](#) and [6](#)

$$a) \quad 0.44(2422.15) = (0.85 + (0.0052)(2258))t$$

$$t = 95.99 \text{ s for evap onstr}$$

$$b) \quad -M_{CV}(0) = -\dot{m}_e t$$

$$0.449 = 0.0052 t$$

$$t = 85.96 \text{ s}$$